

# Computational Hydrology and Remote Sensing at Rice University



The *Computational Hydrology and Remote Sensing* research group at [Rice University](https://www.rice.edu/) focuses on the intersection between water resources, food, and climate. Our research aims to support actionable decision-making by improving hydrological information for monitoring and forecasting of hydrological extremes and their impacts at local scales. To this end, we develop scalable computational approaches for hyper-resolution hydrological prediction targeting regional to global scale applications by leveraging advances in satellite remote sensing, land surface modeling, artificial intelligence, big geospatial data, data fusion, and high-performance computing. More information at [www.waterai.earth](http://www.waterai.earth).

Our research group works on multidisciplinary projects in the following areas (and at their intersections) :

1. Advancing terrestrial hydrology through physics-based AI
2. Satellite remote sensing and AI data assimilation
3. Integrated Earth system modeling and climate change impacts
4. Prediction of extreme hydroclimate events (floods, droughts, landslides, wildfires)
5. Quantitative assessments of climate change and human interventions on water scarcity

## Background:

Successful candidates will have a background in geosciences, environmental science and engineering, climate sciences, applied math, physics, scientific deep learning, or related fields. This is an excellent opportunity to develop expertise in land surface modeling, hydrologic prediction, satellite remote sensing, and data science while contributing to cutting-edge research in Earth science.

## Essential Qualifications:

- Strong analytical skills, ability to think critically and solve problems effectively.
- A solid understanding of calculus and numerical methods.
- Strong programming skills (UNIX/Linux, Python, shell scripting) for processing and visualization of simulation and remote sensing data.
- Experience with geospatial datasets (e.g., GeoTIFF, NetCDF, HDF5, Zarr, dask) and data processing using Python geospatial libraries (e.g., xarray, cartopy, rasterio).
- Experience with machine learning and/or deep learning concepts using Python (e.g., PyTorch, TensorFlow)
- Eagerness to acquire new skills and adapt them to cutting-edge research in hydrology and Earth sciences.
- Excellent written and verbal communication.

## Desired Qualifications:

- Demonstrate a research track record of involvement in topics relevant to computational hydrology, remote sensing, or related fields.
- Experience with at least one advanced programming language (e.g., C, C++, Fortran, etc.)
- Hands-on experience with machine learning or deep learning frameworks (e.g., TensorFlow, PyTorch).

- Familiarity with High-performance computing (HPC) systems and/or cloud computing (e.g., AWS, Google Cloud, etc.).
- Familiarity with version control systems for collaborative code development (e.g., Github, Gitlab)

All researchers will benefit from our group's involvement in national and international collaborative projects, and Rice's thriving and expanding programs, such as the [Rice Data Science Initiative](#), [Data to Knowledge Lab](#), and the [Ken Kennedy Institute](#).

We believe that a diverse team enriches our workplace and enhances our impact. We strongly encourage applications from women, individuals from underrepresented minority groups, and all who can contribute to the further diversification of ideas and perspectives.

### Application for PhD students:

PhD students should submit an application to [Earth, Environmental, and Planetary Sciences](#) (deadline of **October 30th 2026 for the Spring 2027** and **January 8th 2027 for the Fall 2027**). International students should also meet the [language proficiency requirements](#). Prospective graduate students can email Dr. Vergopalan (vergopalan@rice.edu) with the subject "Prospective PhD student" before applying. In the email, please include your unofficial transcripts and curriculum vitae, along with the names and contact information of three references, all in PDF format (not zipped). In your email, include *a brief personal statement explaining why you fit and why you would like to join the group*. We greatly appreciate all the applications, but due to the high volume of submissions, please note that only shortlisted candidates will be notified. Compensation: \$37K/year stipend with benefits plus full tuition (\$66K/year).

### Postdoctoral Researchers:

Postdoctoral positions are currently available in Dr. Vergopalan's group, subject to funding. Outstanding applicants may also wish to explore fellowship opportunities, including the [Pan Postdoctoral Research Fellowship](#) (deadline on **October 25th, 2026**), the [UCAR Climate & Global Change Postdoctoral Program](#) (deadline on **October 16th, 2026**), among others. Applicants should have demonstrated expertise in process-based modeling, remote sensing, or land surface modeling, along with strong programming skills and experience in HPC or cloud computing. Excellent written and verbal communication skills, demonstrated through a strong publication record, are expected. To apply for a position in the group, email a CV, unofficial transcripts, a brief statement explaining your interest, and contact information for three references to Dr. Vergopalan (vergopalan@rice.edu) with the subject line "Prospective Postdoctoral Researcher." We greatly appreciate all the applications, but due to the high volume of submissions, please note that only shortlisted candidates will be notified. Appointments are renewed annually based on performance. Salary is commensurate with experience, ranging from \$60k–70k per year, plus benefits.